

LANGENOIS, Austria

WMO: 110750

Lat: 48.472N

Lon: 15.697E

Elev: 209

StdP: 98.84

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.3	-14.2	1.1	-10.2	-11.5	1.4	-7.3	8.5	6.3	7.3	5.9	2.2	330	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.7	21.5	29.8	20.6	28.0	19.6	22.3	29.7	21.3	28.2	20.4	26.5	2.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.0	25.8	19.0	14.1	24.5	18.1	13.4	23.7	66.5	29.9	62.8	28.5	59.5	26.5	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.9	5.0	-13.7	34.8	3.9	1.9	-16.5	36.2	-18.8	37.3	-21.0	38.4	-23.9	39.7		
(5)			-14.1	24.0	3.7	0.8	-16.7	24.6	-18.9	25.1	-20.9	25.6	-23.6	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-0.2	1.6	5.6	10.9	15.2	18.8	20.6	20.2	15.3	10.1	5.3	0.8	(6)
(7)		DBStd	8.30	4.32	4.40	4.10	4.11	3.46	3.56	3.19	3.30	3.28	3.71	3.70	4.02	(7)
(8)		HDD10.0	1218	317	236	144	39	2	0	0	0	1	46	146	286	(8)
(9)		HDD18.3	3131	575	469	395	225	107	37	14	18	101	254	392	544	(9)
(10)		CDD10.0	1367	1	1	8	65	165	265	329	318	161	50	4	0	(10)
(11)		CDD18.3	238	0	0	0	1	11	52	86	77	11	0	0	0	(11)
(12)		CDH23.3	2232	0	0	0	26	129	498	798	692	86	3	0	0	(12)
(13)		CDH26.7	691	0	0	0	3	18	147	274	236	12	0	0	0	(13)
(14)	Wind	WSAvg	2.1	2.1	2.2	2.5	2.5	2.2	2.1	2.1	1.8	1.9	1.8	1.7	1.7	(14)
(15)	Precipitation	PrecAvg	613	27	22	37	39	74	89	85	82	56	42	34	28	(15)
(16)		PrecMax	875	60	52	97	92	144	236	175	257	146	94	71	54	(16)
(17)		PrecMin	402	3	2	5	0	17	42	29	18	15	5	0	4	(17)
(18)		PrecStd	110	14	14	22	24	35	40	39	49	28	26	15	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.1	20.1	26.3	28.7	32.6	33.9	34.2	28.3	22.3	17.6	12.3	(19)
(20)			MCWB	8.5	9.1	11.9	15.7	18.8	21.6	21.8	21.5	19.2	15.6	12.4	8.8	(20)
(21)		2%	DB	10.2	12.3	17.2	23.1	26.5	30.2	31.6	31.6	25.8	19.7	14.5	9.8	(21)
(22)			MCWB	6.8	8.0	10.6	14.6	18.1	21.2	21.6	21.2	18.3	14.6	11.0	6.9	(22)
(23)		5%	DB	8.0	9.9	14.5	20.7	24.7	28.2	29.7	29.2	23.6	17.7	12.2	7.7	(23)
(24)			MCWB	5.1	6.4	9.3	13.3	17.0	19.7	20.5	20.4	17.3	13.8	9.8	5.3	(24)
(25)		10%	DB	6.0	7.8	12.2	18.6	22.6	26.3	27.8	27.3	21.4	15.8	10.4	6.1	(25)
(26)			MCWB	3.7	4.9	8.0	12.3	15.7	18.9	19.7	19.4	16.1	12.8	8.4	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.7	12.6	16.8	20.0	23.2	23.8	23.6	20.1	16.8	13.0	9.1	(27)
(28)			MCD B	12.7	13.9	18.9	24.8	27.0	30.6	31.5	30.9	26.4	20.4	16.2	11.9	(28)
(29)		2%	WB	7.0	8.1	11.0	15.1	18.7	21.8	22.3	22.1	18.9	15.5	11.6	7.1	(29)
(30)			MCD B	10.1	11.9	16.3	22.2	25.2	28.7	29.8	29.5	24.5	18.6	14.2	9.4	(30)
(31)		5%	WB	5.4	6.6	9.7	13.8	17.6	20.7	21.3	21.0	17.9	14.3	10.0	5.7	(31)
(32)			MCD B	7.6	9.7	13.8	19.7	23.4	26.7	28.1	28.1	22.5	16.9	12.0	7.3	(32)
(33)		10%	WB	4.0	5.1	8.4	12.7	16.5	19.5	20.3	20.0	16.8	13.2	8.7	4.5	(33)
(34)			MCD B	5.9	7.4	11.9	17.9	21.5	25.0	26.4	25.8	20.5	15.6	10.2	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	7.3	9.1	10.9	10.7	10.9	11.1	11.3	9.8	8.4	5.9	5.4	(35)
(36)			MCDBR	8.9	11.4	13.9	14.8	14.4	14.2	14.6	14.7	13.1	11.4	9.4	8.8	(36)
(37)			MCWBR	6.2	7.3	8.0	7.3	6.6	6.1	5.8	6.0	6.2	6.7	6.2	6.3	(37)
(38)		5% WB	MCDBR	8.5	11.0	13.0	13.7	13.2	13.2	13.1	13.8	12.2	10.2	8.8	8.3	(38)
(39)			MCWBR	6.2	7.3	7.7	7.0	6.3	6.0	5.7	6.0	6.1	6.3	6.0	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.335	0.376	0.419	0.420	0.425	0.432	0.429	0.400	0.369	0.341	0.310	(40)	
(41)		taud	2.386	2.348	2.279	2.193	2.232	2.255	2.237	2.262	2.310	2.373	2.405	2.423	(41)	
(42)		Ebn at Noon	752	813	832	829	843	839	826	810	797	761	705	705	(42)	
(43)		Edh at Noon	68	88	111	134	134	132	132	124	107	86	67	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.99	1.86	2.94	4.34	5.04	5.50	5.31	4.74	3.38	2.09	1.06	0.78	(44)	
(45)		RadStd	0.15	0.24	0.38	0.51	0.58	0.47	0.52	0.47	0.43	0.24	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (92 neighbors)		+0.52	N/A	N/A	N/A	+0.39	+0.33	-120	-151	+73	+33

Nomenclature: See separate page